SAMPLE PAPER (made up for testing - not a real exam)

End Semester Examination December 2024

Course: NLP and GenAI

Answer any five questions. All questions carry equal marks.

1. Explain heuristic, machine learning and deep learning approaches to NLP problems. (10 marks)
2. (a) Describe the encoder-decoder architecture of the Transformer. (5 marks)

(b) Why is positional encoding needed in Transformers? (5 marks)

3. What are Large Language Models? Why are they called "large"? (10 marks)
4. Explain zero-shot and few-shot prompting with examples. (10 marks)
5. Explain the steps involved in fine-tuning a pre-trained LLM. List its disadvantages. (10 marks)
6. Explain different types of memory available in LangChain. (10 marks)
7. Compare RNN and LSTM. (10 marks)